Unit: Unit 3: Functions
Topic: Function Transformations

Name: ____________
Date: ____________

Homework (Habanero)

Q1. A patient's dosage d of a medication is modeled by $d(x) = 2x^2 - 5x + 1$. If the dosage is increased by 3, the new function is $d(x) =$
(A) $2x^2 - 5x + 4$
(B) $2x^2 - 5x + 1 + 3x$
(C) $2x^2 - 5x + 1 - 3x$
(D) $2x^2 - 5x - 2$
(E) $5x^2 - 10x + 3$

Q2. The growth of a cell g can be represented by $g(t) = 3t^2$. If the function is horizontally shifted by 2 units to the right, the new equation is $g(t) =$
(A) $3(t-2)^2$
(B) $3(t+2)^2$
(C) $3(t-1)^2$
(D) $9t^2$
(E) $t^2 + 2$

Q3. The function $h(x) = 2x^2 - 1$ represents the spread of a disease. If the function is reflected across the x-axis, the new equation is $h(x) =$
(A) $-2x^2 - 1$
(B) $-2x^2 + 1$
(C) $2x^2 + 1$
(D) $2x^2 - 2$
(E) $x^2 - 2$

Q4. The probability p of a successful medical trial is given by $p(x) = x^2 - 4x + 3$. If the function is vertically shifted up by 2, the new equation is $p(x) =$
(A) $x^2 - 4x + 5$
(B) $x^2 - 4x + 1$
(C) $x^2 + 2x - 3$
(D) $x^2 - 4x - 1$
(E) $2x^2 - 8x + 6$

Q5. The concentration c of a substance is modeled by $c(t) = 4t^2 - 5t$. If the function is horizontally compressed by a factor of 2, the new equation is $c(t) =$
(A) $4(2t)^2 - 5(2t)$
(B) $4(2t)^2 + 5(2t)$
(C) $16t^2 - 10t$
(D) $4t^2 - 5t$
(E) $t^2 - 2t + 1$

Q6. The spread of a disease s can be represented by $s(x) = x^2 - 2x - 3$. If the function is reflected across the y-axis, the new equation is $s(x) =$
(A) $(-x)^2 - 2(-x) - 3$
(B) $x^2 + 2x + 3$
(C) $x^2 - 2x + 3$
(D) $x^2 + 2x - 3$
(E) $-x^2 - 2x - 3$

Q7. A medical experiment's results r can be modeled by $r(t) = 2t^2 + 3t - 1$. If the function is vertically compressed by a factor of 2, the new equation is $r(t) =$
(A) $t^2 +$
rac32t -
rac12
(A) $t^2 -$
rac32t +
rac12
(A) $2t^2 + 3t - 2$
(B) $4t^2 + 6t - 2$
(C) $t^2 - 2t - 1$

Q8. The function $f(x) = x^2 - 2x + 1$ represents the dosage of a medication. If the function is shifted to the left by 1 unit, the new equation is $f(x) =$
(A) $(x+1)^2 - 2(x+1) + 1$
(B) $(x-1)^2 - 2(x-1) + 1$
(C) $x^2 + 2x + 1$
(D) $x^2 - 2x - 1$
(E) $x^2 + 2x - 1$

Open Ended Questions (FRQ)

Q9. Consider the function $g(t) = t^2 - 4t + 3$ representing the growth rate of a cell. Describe and apply a transformation to $g(t)$ such that the new function $g'(t)$ has its vertex at $(2, 0)$. Show all work and explain your reasoning.

Q10. The function $h(x) = 2x^2 - 5x + 1$ models the spread of a disease. Apply a horizontal compression by a factor of 3 and then reflect the resulting function across the x-axis. Write the equation of the final transformed function $h'(x)$ and compare its graph to the parent function $h(x)$. Show all work and explain your reasoning.

Chef's Tips

- Read each question carefully and choose the correct answer
- Use the provided space for free response questions
- Show all work and explain your reasoning for free response questions